

Coffee Sales Analysis Dashboard

Power BI & Excel Project | Business Insight & Data Analysis

Summary

This project analyzes coffee shop sales using Excel and Power BI to identify key demand patterns, revenue drivers, and operational opportunities. The analysis highlights peak-hour concentration, off-peak inefficiencies, and product-level performance, leading to data-driven recommendations aimed at improving revenue consistency, operational efficiency, and overall business performance.

Keywords: Coffee Sales Analysis, Power BI, Data Visualization, Peak Hour Analysis, Revenue Optimization

PROJECT OVERVIEW

This project analyzes coffee shop sales data using both Excel (pivot tables) and Power BI dashboards to identify sales patterns, customer behavior, and revenue opportunities. The goal is to transform raw transactional data into actionable business insights to support data-driven decision-making.

DATASET & TOOLS

Dataset:

<https://www.kaggle.com/datasets/anassarfraz13/coffee-sales-dataset>

Tools Used:

- Microsoft Excel (Pivot Tables)
- Power BI (Dashboard & Visualization)

Techniques:

- Data aggregation (Pivot Table)
- Data modeling
- Time-based analysis
- Data visualization

EXCEL ANALYSIS

		Years(Date)	(All)	
Row Labels	Sum of money			
2024	\$ 82.644,80			
2025	\$ 29.600,78			
Grand Total	\$ 112.245,58			
Row Labels	Sum of money	Count of coffee_name		
3	\$ 15.891,64	494		
10	\$ 13.891,16	426		
2	\$ 13.215,48	423		
9	\$ 9.988,64	344		
11	\$ 8.590,54	259		
12	\$ 8.237,74	259		
5	\$ 8.164,42	241		
6	\$ 7.617,76	223		
8	\$ 7.613,84	272		
7	\$ 6.915,94	237		
1	\$ 6.398,86	201		
4	\$ 5.719,56	168		
Grand Total	\$ 112.245,58	3547		

Weekdaysort	(All)									
Count of coffee_name	Column Labels									
Row Labels	Americano	Americano with Milk	Cappuccino	Cocoa	Cortado	Espresso	Hot Chocolate	Latte	Grand Total	
Afternoon	233	239	164	75	88	56	80	270	1205	
Morning	219	331	122	58	143	44	49	215	1181	
Night	112	239	200	106	56	29	147	272	1161	
Grand Total	564	809	486	239	287	129	276	757	3547	
Row Labels	Count of coffee_name	Sum of money	First Price							
Americano with Milk	809	\$ 24.751,12	33,8							
Latte	757	\$ 26.875,30	38,7							
Americano	564	\$ 14.650,26	28,9							
Cappuccino	486	\$ 17.439,14	38,7							
Cortado	287	\$ 7.384,86	28,9							
Hot Chocolate	276	\$ 9.933,46	38,7							
Cocoa	239	\$ 8.521,16	38,7							
Espresso	129	\$ 2.690,28	24							
Grand Total	3547	\$ 112.245,58								

Description:

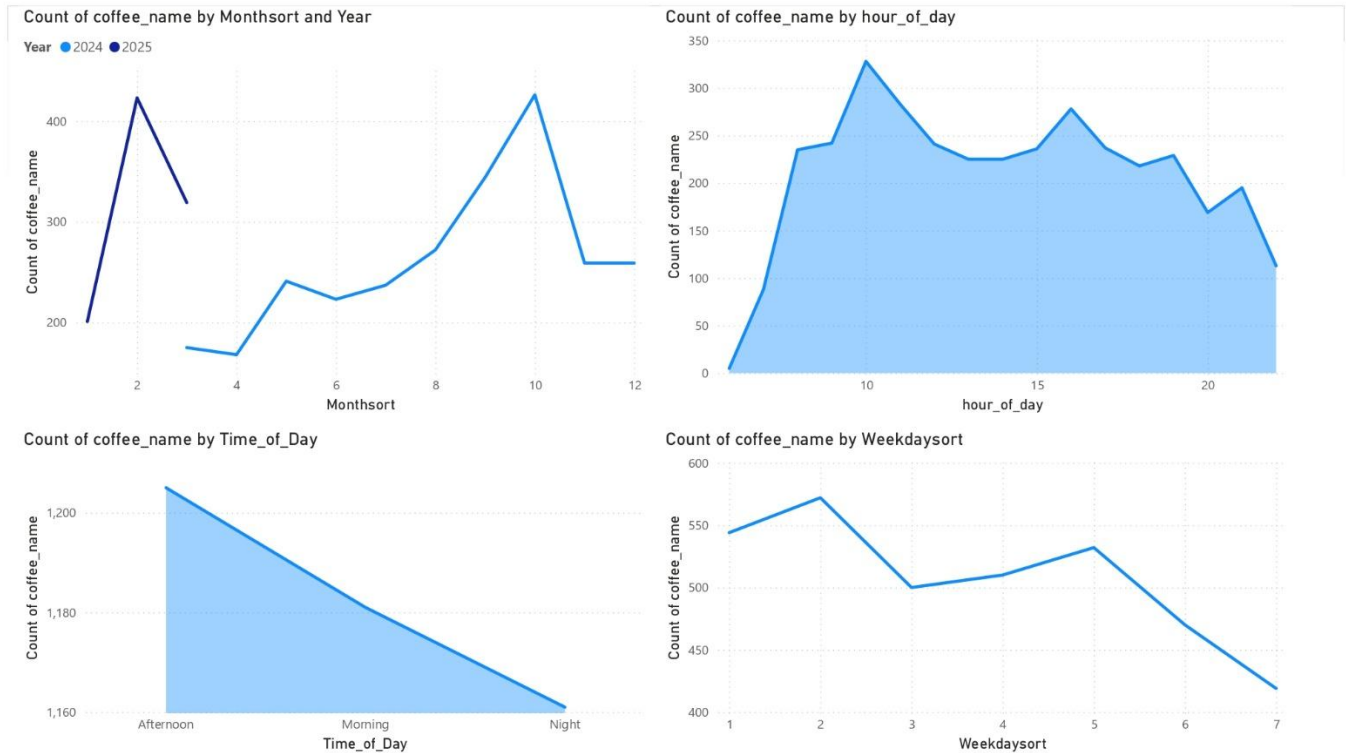
Excel was used to perform initial data aggregation and analysis using pivot tables, including revenue by year, product performance, and time-based sales breakdown.

Key Findings from Excel:

- Total revenue: **\$112,245.58**
- Highest revenue year: **2024 (\$82,644.80)**

- Best-selling products:
 - Americano with Milk (809 orders)
 - Latte (757 orders)
- Lowest performing product:
 - Espresso (129 orders)

POWER BI DASHBOARD



Description:

An interactive dashboard was built in Power BI to visualize sales trends by hour, weekday, and time of day, enabling deeper business insights and pattern recognition.

KEY INSIGHTS

1. Sales are highly concentrated during peak hours (09:00–11:00), indicating a critical revenue window where demand is at its highest and operational efficiency becomes essential to capture maximum sales.
2. A significant drop in sales occurs during evening and night periods, highlighting underutilized business capacity and missed revenue opportunities during off-peak hours.
3. Product demand is heavily concentrated on a few top-performing items (Americano with Milk and Latte), suggesting a strong customer preference pattern that drives the majority of sales.

BUSINESS RECOMMENDATIONS

1. Optimize staffing allocation and service speed during peak hours (09:00–11:00) to reduce queue time and prevent lost sales, as aligning workforce with demand is critical for maximizing revenue during high-traffic periods (Lightspeed, 2023).
2. Activate off-peak demand by implementing time-based promotions such as evening discounts (e.g., 15–20% after 6 PM), product bundling, or happy hour campaigns to increase customer traffic and improve revenue utilization (Square, 2023).
3. Leverage high-performing products by prioritizing them in promotions, upselling strategies, and menu placement to maximize revenue contribution and improve overall sales efficiency.

BUSINESS IMPACT

By optimizing peak-hour operations and activating off-peak demand, the business can improve revenue consistency, maximize daily

sales potential, and increase operational efficiency. These strategies enable better utilization of resources while enhancing customer experience and overall profitability

CONCLUSION

This project demonstrates how combining Excel and Power BI can provide both detailed data analysis and interactive visualization. By leveraging these insights, businesses can optimize operations, improve customer experience, and increase overall revenue performance.

REFERENCES

Lightspeed. (2023). *Using the Sales by Hour of Day Report*
<https://x-series-support.lightspeedhq.com/hc/en-us/articles/25534114054683>

Square. (2023). *How to Attract Customers During Off-Peak Times*
<https://squareup.com/gb/en/the-bottom-line/reaching-customers-off-peak>